Iman Rahmaty

Email: imanrht@gmail.com Github: https://github.com/ImanRHT Linkedin: linkedin.com/in/iman-rahmaty

Research Interests: Distributed Systems, Multi-Agent Reinforcement Learning (RL), Federated and Collaborative Learning, Network Performance Optimization, Wireless and Mobile Networks, IoT, Fog/Edge Computing Systems

EDUCATION

MSc. Computer Engineering/Networking Sharif University of Technology (SUT), Tehran, Iran
Graduated Sep 2022, 3.5/4 GPA

Thesis Title: A Resource Allocation Algorithm based on Deep RL (DRL) for Mobile Edge Computing (MEC).
Supervisor: Prof. Ali Movaghar

BSc. Industrial Engineering Khajeh Nasir Toosi University of Technology (KNTU), Tehran, Iran
Graduated Sep 2019

High School Diploma in Mathematics and Physics Kamal High School, Tehran, Iran
Graduated Jun 2013

ACADEMIC EXPERIENCE

Research Engineer at EdgeAI Lab 2023-Present
Supervisor: Prof. Hamed Shah-Mansouri Department of Electrical Engineering, SUT

Research Theme: Developing hierarchical multi-agent RL-based approaches for collaborative offloading decision-making in heterogeneous MEC systems, focused on centralized training and decentralized execution.

Research Assistant at Performance and Dependability Lab (PDL) 2019-2022
Supervisor: Prof. Ali Movaghar Department of Computer Engineering, SUT

Research Theme: Developing DRL-based computation offloading algorithms for MEC to optimize distributed decision-making and maximize end-user QoE for time-sensitive and energy-constrained tasks.

Teaching Assistants

- **Performance Evaluation of Computer Systems** (Head TA) SUT, 2020-2022
  Prof. Ali Movaghar and Dr. Mahdi Dolati
- **Software Defined Networking** (Head TA) SUT, 2022
  Prof. Ali Movaghar and Dr. Mohammad Hosseini
- **Verification of Reactive Systems** and **Theory of Machines and Languages** (Head TA) SUT, 2021
  Prof. Ali Movaghar
- **Wireless Networking** SUT, 2021
  Prof. Ali Mohammad Afshin Hemmatyar

Reviewer, IEEE Internet of Things Journal IEEE, 2025

PUBLICATION

- I. Rahmaty, H. Shah-Mansouri, A. Movaghar, "QECO: A QoE-Oriented Computation Offloading Algorithm based on Deep Reinforcement Learning for Mobile Edge Computing", *IEEE Transactions on Network Science and Engineering*, vol. 12, no. 4, pp. 3118-3130, July-Aug. 2025.
- I. Rahmaty, H. Shah-Mansouri, M. Hosseini, A. Movaghar, "A Collaborative Computation Offloading Framework based on Multi-Agent Deep Reinforcement Learning in Partially Observable Mobile Edge Computing Environments", *Work in progress.*

HONORS

- ❖ Ranked in the top 10% of M.Sc. students in the Dept. of Computer Engineering at SUT (Class of 2019) 2022
- ❖ Ranked 55th among 60,000 in the nationwide M.Sc. Computer Engineering entrance exam. 2019
- ❖ Ranked Top 1% among 180,000 in the nationwide B.Sc. entrance exam in Mathematics and Physics. 2014
- ❖ Achieving the 3rd position in the RoboCup Competition (IranOpen) 2012

SELECTED COURSES

- Computer Performance Evaluation $\mathcal{A}^+$
- Verification of Reactive Systems $\mathcal{A}^+$
- Advanced Network Security $\mathcal{A}^+$
- Wireless Networking $\mathcal{A}^+$
- Theory of Distributed Systems $\mathcal{A}$
- Computer Network $\mathcal{A}$
- IT Enterprise architecture $\mathcal{A}^+$
- Computer Network Management $\mathcal{B}^+$

SELECTED PROJECTS

- **Hierarchical Cooperative Multi-Agent D3QN** EdgeAI, 2025
  Developed a multi-agent DRL framework for cooperative offloading decision-making based on Dec-POMDP.
- **Collaborative Cloud-Edge-End MEC Environment** EdgeAI, 2024
  Event-based simulation of multi-layer collaborative offloading processes across cloud, edge nodes, and end devices.
- **Dueling Double Deep Q-Networks (D3QN)** EdgeAI, 2023
  Developed a distributed DRL framework for offloading decision-making under uncertainty based on MDPs.
- **Resource-constrained MEC Environment** PDL, 2021
  Time-based simulation of computation and communication processes in resource-constrained MEC systems.
- **Queueing System** SUT, 2020
  Event-based simulation and performance evaluation of M/M/1/K queues with various service disciplines.

SKILLS

- **General:** Wireless Networks, MEC, Multi-Agent RL, Simulation and Modeling, Performance Evaluation
- **Programming Languages:** Python, R, Bash, C++, SQL
- **Machine Learning:** TensorFlow, PyTorch, Keras, Scikit-learn
- **Data Analysis:** Pandas, NumPy, Matplotlib, Jupyter Notebooks
- **Frameworks & Tools:** Linux, Mininet, Ns-3, Git, LaTeX, Vim, Flask, Visio
- **Language Proficiency:** Persian (Native), English (Working proficiency)
  - TOEFL (IBT) Score: 108/120 (R: 30, L: 28, S: 22, W: 28)

CERTIFICATION

Interactive Learning Tehran Institute for Advanced Studies (TeIAS), 2021
Certification of Completion in Deep Reinforcement Learning Course, Inst: Prof. Majid Nili
Machine Learning and Deep Learning in Python Start-Tech Academy, 2020
Certification of Completion in Udemy Online Course
Data Science Tose'e Higher Education Institute, 2019
Certification of Completion in Data Science Course, Inst: Dr. Yaser Zerehsaz
Advanced Python Topics Remis Arjang Institute, 2018
Certification of Completion in Advanced Python Course, Inst: Dr. Peyman Hooshmandi
LPIC1 Anisa Iran Linux House, 2017
Certification of Completion in Linux Administrator Course, Inst: Dr. Amir Abbasi

REFERENCES

Prof. Ali Movaghar movaghar@sharif.edu
Professor of Computer Science and Engineering, SUT, Tehran, Iran.
Prof. Hamed Shah-Mansouri hamedsh@sharif.edu
Assistant Professor of Electrical Engineering, SUT, Tehran, Iran.
Prof. Mohammad Hosseini m-hosseini@sbu.ac.ir
Assistant Professor of Computer Engineering, Shahid Beheshti University, Tehran, Iran.